# Aiden Zhang

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## Technical Skills

**Languages:** Java, Python, C++, SQL, HTML, CSS, TypeScript, JavaScript
**Frameworks:** React.js, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, ROS 2, pandas, numpy, Spring Boot
**Developer Tools:** Git, Github, Docker, VSCode, Foxglove, PyCharm, Eclipse, Vite, Chrome DevTools

## Projects

**Computer Vision Food Recognition and Nutritional Analysis System**

*Python, PyTorch, YOLO, OpenCV, NumPy, pandas, TensorFlow* June 2025 - October 2026

- Built a real-time food detection app that finds and identifies 5 food categories in live webcam video, replacing an earlier TensorFlow classifier (96% test accuracy) with a fine-tuned YOLO11 model that boxes each food and detects several items at once
- Trained the model on **245 labeled images, reaching 98% mAP50** on a validation set averaging ~13.5 FPS across 300 frames
- Ran a 5-month pilot with **6 peers and family members,** including a **visually impaired participant** who used TTS to access nutrition feedback supporting nutrition-aware food choices; participants reported healthier eating habits and an average **1.1% reduction in body-fat percentage**

**Recall - AI Study Assistant Chrome Extension**

*Java 21, Spring Boot, React, TypeScript, PostgreSQL, SQL, Vite, CSS, Git, npm, Vite* October 2026 - Present

- Built a chrome Manifest V3 side-panel extension with React and TypeScript that generates fill-in-the-blank quizzes from pasted or uploaded course notes with Hints On/Off modes & retry of missed questions
- Designed a Java 21 and Spring Boot REST API backed by PostgreSQL that persists quizzes, questions and attempts across sessions

**Autonomous Robot Navigation & LiDAR Mapping - WATonomous**

*C++, ROS2, Foxglove, Docker* Sept 2026

- Built a 4-node autonomous navigation pipeline in C++/ROS 2, integrating LiDAR mapping, odometry-based map fusion, A* pathfinding, and 10Hz Pure Pursuit control
- Engineered 100 x 100 m occupancy grids with 0.1m resolution, 0.65 m obstacle inflation, and 8-direction A* planning

**Drone Behaviour Tree Telemetry System**

*Python, ROS2, MAVLINK, py_trees, pymavLink, Git* Sept 2026 - Oct 2026

- Implemented an initial ROS 2 behaviour-tree telemetry protocol, subscribing to live airside snapshots and automatically mapping 100% of observed behaviors to compact IDs and parent-child relationships for bandwidth-constrained MAVLink STATUSTEXT
- Developed an air-to-ground telemetry pipeline to encode active decision paths and reconstruct autonomy state groundside, targeting ≤ 50-byte STATUSTEXT payloads for debugging without LTE connectivity, which was tested later in ArduPilot

**Bedside Patient Monitoring System**

*Arduino UNO, C++, Embedded Programming, I2C, UART, Serial Communication, ADC* October 2026 - Present

- Developed an embedded monitoring prototype integrating heart-rate sensing & 3-axis acceleration measurements to track pulse
- Implemented non-blocking sensor acquisition, signal filtering, and configurable LED alerts for activity changes which are tracked

## Experience

**Camp Counsellor -** *City of Richmond Hill* May 2025 - August 2026

- Led daily programs for children ages 4-12, working with staff and parents to resolve issues

## Certifications

**Stanford AI Machine Learning Specialization** July 2025 - October 2025

*PyCharm, PyTorch, Python, TensorFlow, Scikit-learn, NumPy, Langchain, LLM*

- Built ML models using neural networks, regression, decision trees, clustering, and recommender systems

## Education

**University of Waterloo -** *BASc in Computer Engineering with Co-op* Sept 2026 - April 2031

- **Coursework:** C++ Programming, Physics, Linear Algebra, Calculus, Engineering Prototype and Design